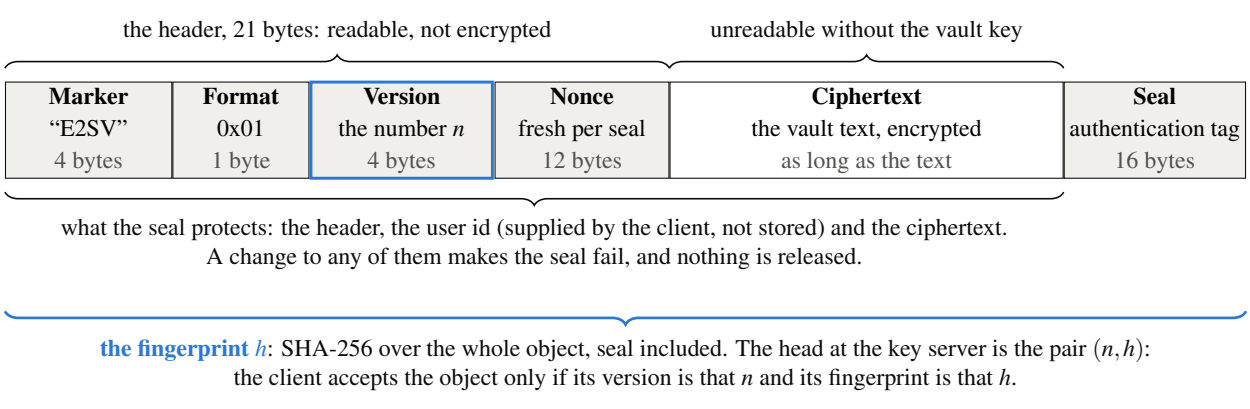
the header, 21 bytes: readable, not encrypted
unreadable without the vault key
Marker
“E2SV”
4 bytes
Format
0x01
1 byte
Version
the number n
4 bytes
Nonce
fresh per seal
12 bytes
Ciphertext
the vault text, encrypted
as long as the text
Seal
authentication tag
16 bytes
what the seal protects: the header, the user id (supplied by the client, not stored) and the ciphertext.
A change to any of them makes the seal fail, and nothing is released.
the fingerprint h: SHA-256 over the whole object, seal included. The head at the key server is the pair (n, h):
the client accepts the object only if its version is that n and its fingerprint is that h.